<u>Class Test 08</u>

<u>Instructions:</u>

1. **Write a Summary (Softcopy, PDF Format) on ExceptionHandlingTutorial_Spring2024-2025 and DatabaseConnectionTutorial.**
2. **Submit pdf in link provided in your VUES Student Account**
3. **The name of PDF must be your ID**

<u>Start from here:</u>⟶ From the slide ExceptionHandlingTutorial_Spring2024-2025, We get to know that an exception is an identifier raised during PL/SQL execution. It occurs when an Oracle error happens. It is raised explicitly by the developer. To handle exceptions, we have to trap the exception using handlers. Also propagate the exception to the calling environment. There are 3 types of exceptions:

1. Predefined Oracle Server Exceptions: It uses Oracle's built-in names such as NO_DATA_FOUND, TOO_MANY_ROWS, INVALID_CURSOR, ZERO_DIVIDE, DUP_VAL_ON_INDEX
2. Non-predefined Oracle Server Exceptions: Must be declared and associated with specific Oracle error codes using PRAGMA EXCEPTION_INIT.
3. User-defined Exceptions: Declared in the declarative section and explicitly raised using the RAISE statement.

Trapping Exceptions Syntax:

```
EXCEPTION
  WHEN exception1 [OR exception2 ...] THEN
      statements
  WHEN OTHERS THEN
      default handler
```

There are two functions for trapping exceptions:

1. SQLCODE: Returns numeric error code.
2. SQLERRM: Returns associated error message.

Example:

```
WHEN OTHERS THEN
  v_error_code := SQLCODE;
  v_error_message := SQLERRM;
  INSERT INTO errors VALUES (v_error_code, v_error_message);
END;
```

Exceptions can be handled in different calling environments, including SQL*Plus, Procedure Builder, Oracle Developer Forms, Precompiler applications, An enclosing PL/SQL block etc. Exception propagation allows subblocks to either handle or pass exceptions to an outer block. For custom error messaging, the RAISE_APPLICATION_ERROR procedure is used within stored subprograms to return user-defined error. The syntax is:
raise_application_error(error_number, message [, {TRUE | FALSE}]);

From DatabaseConnectionTutorial slide**,** we get to know about two types of connection,

1. Connecting to Oracle Database from C#:

First of all, we need to declare a connection string in the app.config file. In the connection string, we need to provide necessary information like name of the connection string, data, source, TCP, host name, port number, dedicated server, service name, and database id and password. Then, this connection string will be used where ever we need to interact with database. For using this connection string, we will use OracleConnection class from Oracle.ManageDataAccess.Client library. We will provide the connection string name in oracleConnection class to start the connection. After that, we have to open the connection and write the SQL query to insert data. Next, we will declare a cmd object using OracleCommand class to execute the action. In the OracleCommand, we have to pass the SQL query and connection string object. Then we will be able to insert the data using cmd object. After executing the command we will see that our data will added inside our oracle database.

2. Connecting to Oracle Database from PHP:

For setting up connection with php project, first we need to have xampp installed. Then navigate to php.ini inside the xampp folder. Open php.ini with notepad. Press CTRL + F and search with keyword “oracle”. Remove (;) extension and save the file to enable Oracle 10g for with Apache service for Xampp. After that download OracleInstantClient from the oracle’s official website and extract the files into local disk C: by making a new folder called Oracle. Setup the Environment for InstantClient by copying the path of the folder and setting it in environment paths. Now from location: C:/Oracle/InstantClient_19_25 folder copy oci.dll, Oraociei19.dll, and Oraons.dll files then paste it into location: C:/Xampp/apache/bin. Finally, Open Xampp and start Apache service. Then click on the admin Button. It will direct to Xampp webpage. From there click on PhpInfo button and it will show available, info page where the services are in use. Search by pressing CTRL + F and type "oci8". If it shows 0Ci8 support, 0Ci8 D Trace support, 0Ci8 version info, then the system is ready to use Oracle 10g as its database for the project. In the project write code using oci_connect() function to connect with your oracle database.